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Balancing Approach

To balance the 'OTHER' category itself also make the class as all almost balanced with other classes

- **Determine the target sample size** by taking the minimum sample count across all categories inside 'OTHER' ($n = 811$).
- **Downsample each subcategory within 'Other'** to match this target, preserving smaller subcategories that already have fewer samples.
- **Combine the balanced 'Other' subcategories with the remaining non-Other categories**, keeping their original distribution.

This approach results in a more balanced dataset, reducing the risk of bias toward the "Other" category and improving model generalization across all classes.

With countVectorizer

SVC (with classweight)	Accuracy	Recall	Precision	F1-score
BOW	0.63	0.71	0.59	0.63
POS	0.15	0.19	0.15	0.13
NER	0.18	0.20	0.17	0.12
POS+NER	0.20	0.23	0.19	0.17

With TF-IDFVectorizer

SVC (with classweight)	Accuracy	Recall	Precision	F1-score
BOW	0.66	0.72	0.61	0.65
POS	0.13	0.17	0.14	0.12
NER	0.18	0.20	0.17	0.12
POS+NER	0.20	0.22	0.19	0.16

SVC ()	Accuracy	Recall	Precision	F1-score
BOW	0.67	0.66	0.65	0.65
POS	0.34	0.13	0.15	0.11
NER	0.41	0.14	0.15	0.11
POS+NER	0.40	0.17	0.20	0.15

SVC ()	Accuracy	Recall	Precision	F1-score
BOW	0.684	0.67	0.66	0.66
POS	0.32	0.12	0.20	.10
NER	0.41	0.14	0.15	0.11
POS+NER	0.394	0.17	0.30	0.16

Multinomial Naive Bayes	Accuracy	Recall	Precision	F1-score
BOW	0.64	0.67	0.60	0.62
POS	0.271	0.11	0.15	0.09
NER	0.21	0.10	0.13	0.07
POS+NER	0.39	0.15	0.19	0.14

Multinomial Naive Bayes	Accuracy	Recall	Precision	F1-score
BOW	0.613	0.42	0.75	0.46
POS	0.27	0.11	0.11	0.07
NER	0.22	0.10	0.13	0.06
POS+NER	0.42	0.14	0.19	0.12

MLP	Accuracy	Recall	Precision	F1-score
BOW	0.64	0.70	0.60	0.64
POS	0.32	0.15	0.18	0.13
NER	0.410	0.14	0.15	0.11
POS+NER	0.36	0.18	0.25	0.16

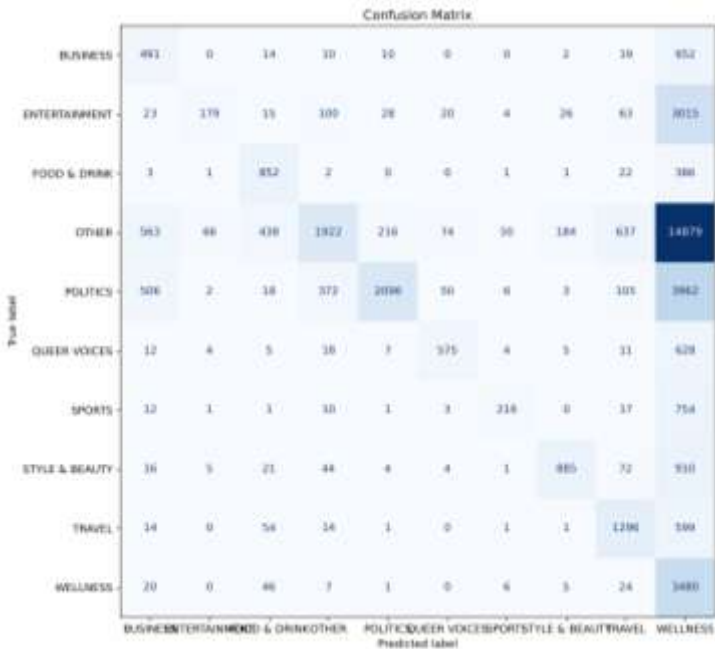
MLP	Accuracy	Recall	Precision	F1-score
BOW	0.550	0.48	0.49	0.48
POS	0.30	0.15	0.20	0.12
NER	0.41	0.14	0.18	0.11
POS+NER	0.36	0.18	0.25	0.16

HMM	Accuracy	Recall	Precision	F1-score
BOW	0.182	0.11	0.11	0.09
POS	0.044	0.09	0.09	0.03
NER	0.041	0.11	0.10	0.03
POS+NER	0.07	0.09	0.10	0.06

HMM	Accuracy	Recall	Precision	F1-score
BOW	0.13	0.10	0.11	0.08
POS	0.0445	0.09	0.09	0.03
NER	0.38	0.10	0.13	0.07
POS+NER	0.07	0.9	0.10	0.6

Confusion Matrix :

1) SVC(with class weights) X BOW_tfidf



2) SVC() X BOW_tfidf



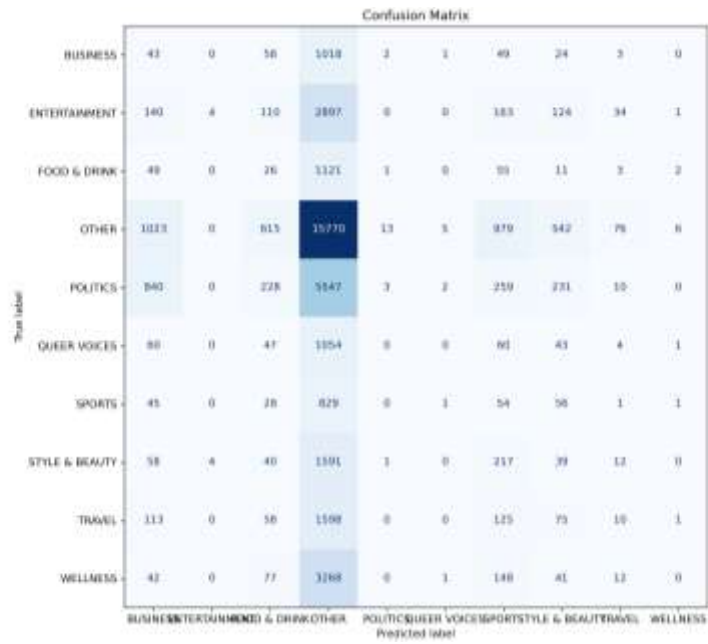
3) NB X BOW



4) MLP X BOW



5)HMM X NER_tfidf



6) HMM X BOW

